

3.1.5 Common Built-in Functions (min, max, len, sum)

len () عناصر گنتا ہے، min () سب سے چھوٹا، max () سب سے بڑا، sum () اعداد کا مجموعہ، اور sorted () نئی ترتیب شدہ

فہر...

CS XII

Chapter 3: Programming Fundamentals

Lecture 06 of 18

3.1.5 Common Built-in Functions (min, max, len, sum)

LEARN IT
Several built-in functions work across lists, tuples, sets and dictionaries. You do not call them with a dot on the object (except when a type also has its own method, such as list .sort()).

Function	Purpose	Works on
min()	Smallest element (numeric or alphabetical)	List, tuple, set, dictionary keys
max()	Largest element	List, tuple, set, dictionary keys
len()	Number of elements (for a dict: number of keys)	List, tuple, set, dictionary
sum()	Adds numeric elements	List, tuple, set, dictionary values via sum(d.values())
sorted()	Returns a new sorted list ; original unchanged	List, tuple, set, dictionary keys

```
s = {5, 1, 9, 3, 7}
print(len(s))
print(min(s))
print(max(s))
print(sum(s))
print(sorted(s))
```

OUTPUT
5
1
9
25
[1, 3, 5, 7, 9]

5 + 1 + 9 + 3 + 7 = 25. sorted(s) always returns a **list**, even when the input is a set.

WORKED EXAMPLE
The same five functions on four structures.

```
nums = [8, 3, 5, 3]
t = (8, 3, 5, 3)
s = {8, 3, 5}
d = {"math": 80, "eng": 70, "cs": 90}
print(len(nums), min(nums), max(nums), sum(nums), sorted(nums))
print(len(t), min(t), max(t), sum(t), sorted(t))
print(len(s), min(s), max(s), sum(s), sorted(s))
print(len(d), min(d), max(d), sum(d.values()), sorted(d))
```

OUTPUT
4 3 8 19 [3, 3, 5, 8]
4 3 8 19 [3, 3, 5, 8]
3 3 8 16 [3, 5, 8]
3 cs math 240 ['cs', 'eng', 'math']

On the dictionary, min(d) and max(d) compare **keys as strings**: "cs" comes first alphabetically, "math" last. sum(d) would try to add the keys "math" + "eng" + ... and fail. Use sum(d.values()) — 80 + 70 + 90 = 240. The set dropped the duplicate 3, so len(s) is 3 and sum(s) is 16.

sorted() versus .sort()

```
original = [8, 3, 5]
print("before sort:", original)
print("sorted():", sorted(original))
print("after sorted():", original)
original.sort()
print("after .sort():", original)
```

OUTPUT
before sort: [8, 3, 5]
sorted(): [3, 5, 8]
after sorted(): [8, 3, 5]
after .sort(): [3, 5, 8]

.sort() exists only on lists and rewrites the original. sorted() works on any iterable and leaves the original alone.

REMEMBER IT
LEN-MIN-MAX-SUM — count, smallest, largest, total. For a dictionary: len / min / max / sorted use **keys**; sum needs .values() .

COMMON MISTAKE
sum() needs numbers. sum(["a", "b"]) raises TypeError . Empty structures: len([]) is 0, but min([]) and max([]) raise ValueError because there is no smallest of nothing.

اردو میں سمجھیں

len() عناصر گنتا ہے، min() سب سے چھوٹا، max() سب سے بڑا، sum() اعداد کا مجموعہ، اور sorted() نئی ترتیب شدہ فہرست دیتا ہے بغیر اصل کو بدلے ہیں۔ ڈکشنری پر min اور max کلیدوں پر چلتے ہیں؛ قدروں کا مجموعہ sum(d.values()) سے نکالیں۔ sort() صرف فہرست کا میٹھڈ ہے جو اصل فہرست بدل دیتا ہے۔

CHECK YOURSELF

1. For s = {5, 1, 9, 3, 7}, what are len(s), min(s), max(s), sum(s) ?

2. Does sorted(s) return a set or a list?

3. How do you total the values of {"math": 80, "eng": 70, "cs": 90} ?

Answers: 1. 5, 1, 9, 25. 2. A list: [1, 3, 5, 7, 9] . 3. sum(d.values()) → 240.

Next lecture: 3.1.6 Activity: Student Attendance System